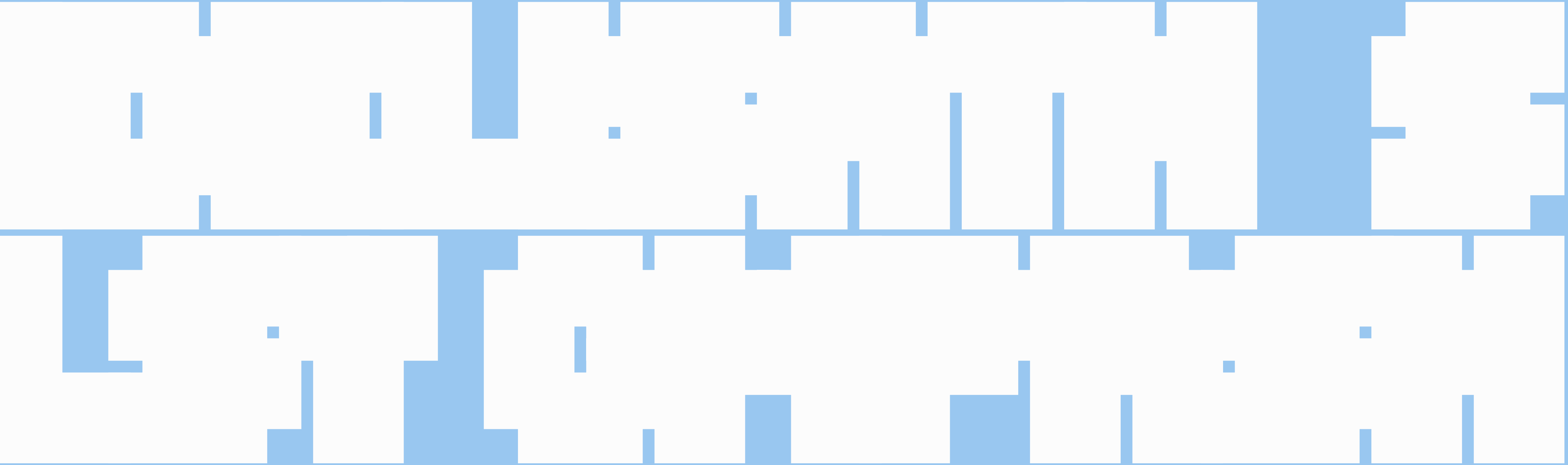


felipe morales carbonell

OUTLINE

i) Idealisation and understanding
ii) Understanding and compression
iii) Compression strategies and schemes
iv) Idealisation as compression
v) Measuring idealisation

IDEALISATION

Different forms of idealisation are common in science:

- Abstract idealisations (formal models)
- Substractive idealisations (simplified models)
- Multiple models (dual models, etc.)

Why are they used? They provide props for dealing with targets of inquiry.

The problem: they are often literally false about their targets (no frictionless surfaces, etc.).

How can falsity serve epistemic purposes?

Now I am not unaware that someone at this point may object that for the purpose of these proofs I am assuming as true the proposition that weights suspended from a balance make right angles with the balance—a proposition that is false, since the weights, directed as they are to the center [of the universe], are convergent. To such objectors I would answer that I cover myself with the protecting wings of the superhuman Archimedes, whose name I never mention without a feeling of awe. For he made this same assumption in his *Quadrature of the Parabola*.[10] And he did so perhaps to show that he was so far ahead of others that he could draw true conclusions even from false assumptions.

Galileo - *De Motu*

IDEALISATION CAN BE HELPFUL FOR UNDERSTANDING

I believe that idealisations can be useful when we aim for different forms of *understanding*.

I take understanding as a non-factive epistemic status.* Understanding a target through a representation does not mean that this representation is true. So in principle idealisations do not present an obstacle to understanding.

* Still, I think that in many kinds of context whether the content of someone's understanding is true is relevant to understanding attributions. The view is complicated. I discuss elements of it in 2021, 2023, 2025.

THE GOODNESS OF IDEALISATION

How, then, is idealisation useful for understanding?

Some answers:

i) Idealisation helps to bootstrap less-idealised models.
ii) Idealisation helps in producing simpler, more usable (computable, tractable, ergonomic, etc.), models.
iii) Idealisation enables appropriate inferences about the phenomena of interest (Suárez 2015, Kuorikoski & Ylikoski 2015, Frigg & Nguyen 2017).
iv) Idealisation singles out causal factors that make substantial differences (Strevens 2008).

THE NEED FOR A DESCRIPTIVE FRAMEWORK

I think the literature captures several aspects of the goodness of idealisations w.r.t. understanding.

However, I think these approaches are descriptively unsatisfactory. My goal here is to present a different framework to describe the function of idealisation.

This framework links understanding and *compression*.

WILKENFELD ON UNDERSTANDING AS COMPRESSION

The idea that understanding and compression are linked is not particularly new, but recent advances in computer science and cognitive science have brought it to the fore in interesting ways.

The first to capture the idea in the recent literature on understanding was Wilkenfeld (2019). According to him, understanding is compression—in particular, understanding of a phenomenon is the possession of a suitably compressed representation of that phenomenon.

Degrees of understanding are a function of the degree of compression: roughly, the more capable to make use of a compressed representation of a phenomenon a subject is, the more they understand of the phenomenon.

UNDERSTANDING-AS-COMPRESSION A person p1 understands object o in context C more than another person p2 in C to the extent that p1 has a representation/process pair that can generate more information of a kind that is useful in C about o (including at least some higher order information about which information is relevant in C) from an accurate, more minimal description length [representation.]

UNDERSTANDING AS PREDICTIVE COMPRESSION

In more recent work, Queloz & Beckmann (ms.) have developed the idea that understanding is competence through predictive compression.

We care about understanding because it predicts robust competence (the ability to achieve aims over a broad range of circumstances). Our concept of understanding has the function of serving as a proxy for this property.

> *[...] understanding is the result of convergent pressures on social agents to predict the world using models that are not only accurate, but also compressed enough to be stored, demonstrated, and transmitted.*

The idea is that robust competence is acquired through the development of compressed models of the targets of interest.

COMPRESSING GRAPHS

I (2023) have developed an account of the format of the kinds of representations that underlie understanding that incorporates an element of compression.

The basic idea is that the format of the representations underlying understanding is graphical rather than symbolic/propositional. To understand is to be in a cognitive state that can be characterized in terms of four elements: a set of internal graphs (the representations the subjects carry), a set of external graphs (the structure of the targets), a mapping between them, and a dispositional profile to act in certain ways w.r.t the other elements of the state.

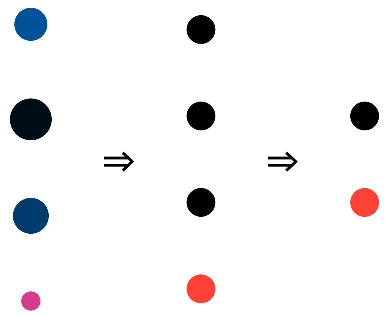
On top of this, we add compression: internal graphs are compressed representations of the external graphs. Degrees of understanding are also a (complex) function of the degree of compression, with some subtleties.

INTERNAL AND EXTERNAL COMPRESSION

We need to distinguish between two senses in which a representation can be compressed.

Supposing that a representation r is semantically linked to a target (t), we can say that r compresses in so far as r holds information about r in a way such that the *length* of r is smaller than t. In this case, we are talking about *external compression*.

A representation r compresses another representation r' if r can be said to hold information that r' has, but its length is smaller than the length of r'. This is *internal compression*. We usually only speak of internal compression when there is a presupposition that either r or r' represents a target t such that $t \neq r \neq r'$ (r could externally compress r').



AN MDL APPROACH

We begin with data about a target. Actually, we begin with an exhaustive description of the data D.

We produce a hypothesis or model of the data, M, and a description of the data given the model $D|M$.

When the lenght of D is less than the lenght of the model plus a description of the data given the model, we have compression:

$$L(D) > L(M) + L(D|M)$$

In an MDL framework we distinguish between *model compression* (minimizes the lenght of M) and *data compression* (minimizes the length of $D|M$).

LOSSY AND LOSSLESS COMPRESSION

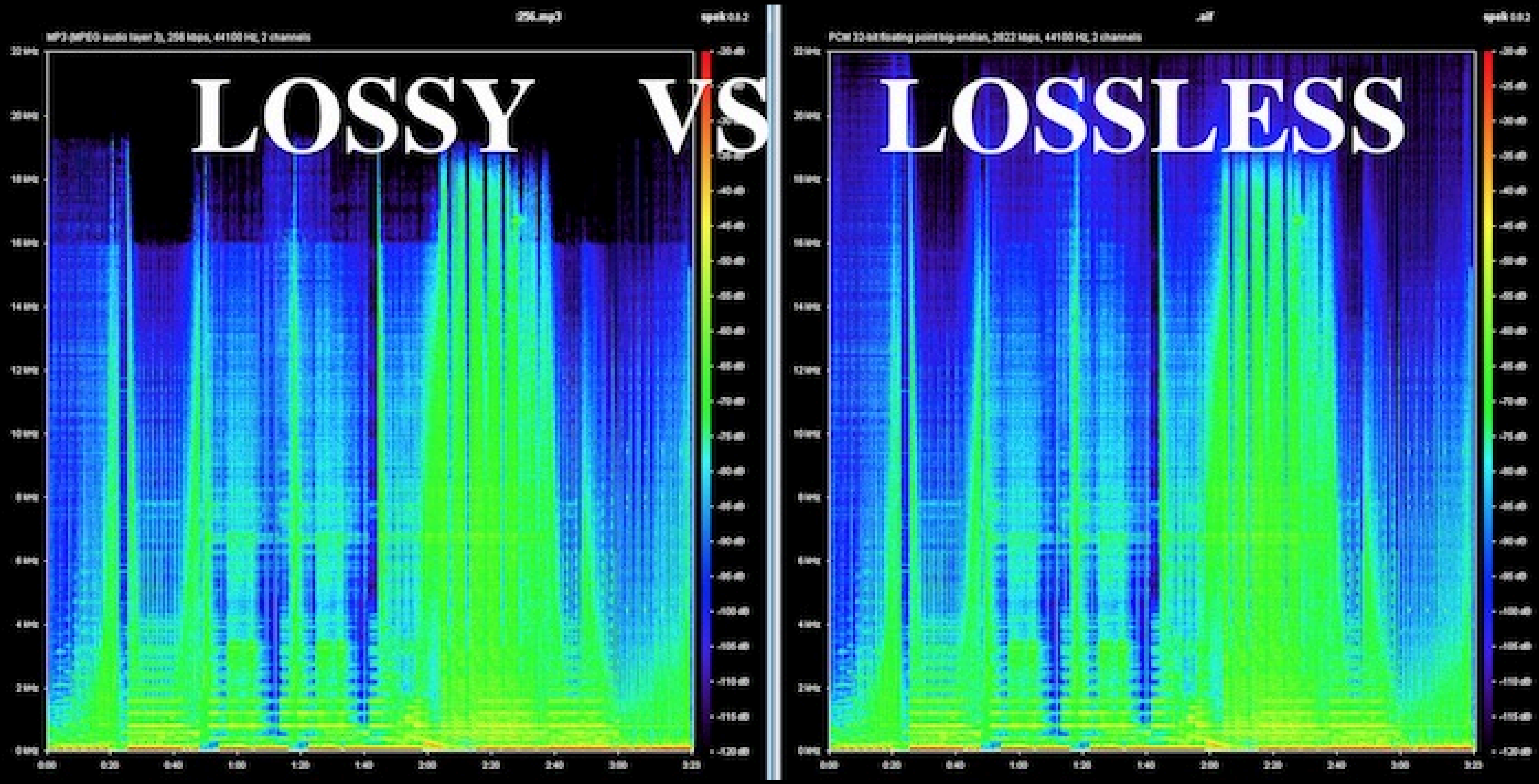
Another key distinction is between *lossless* and *lossy* compression.

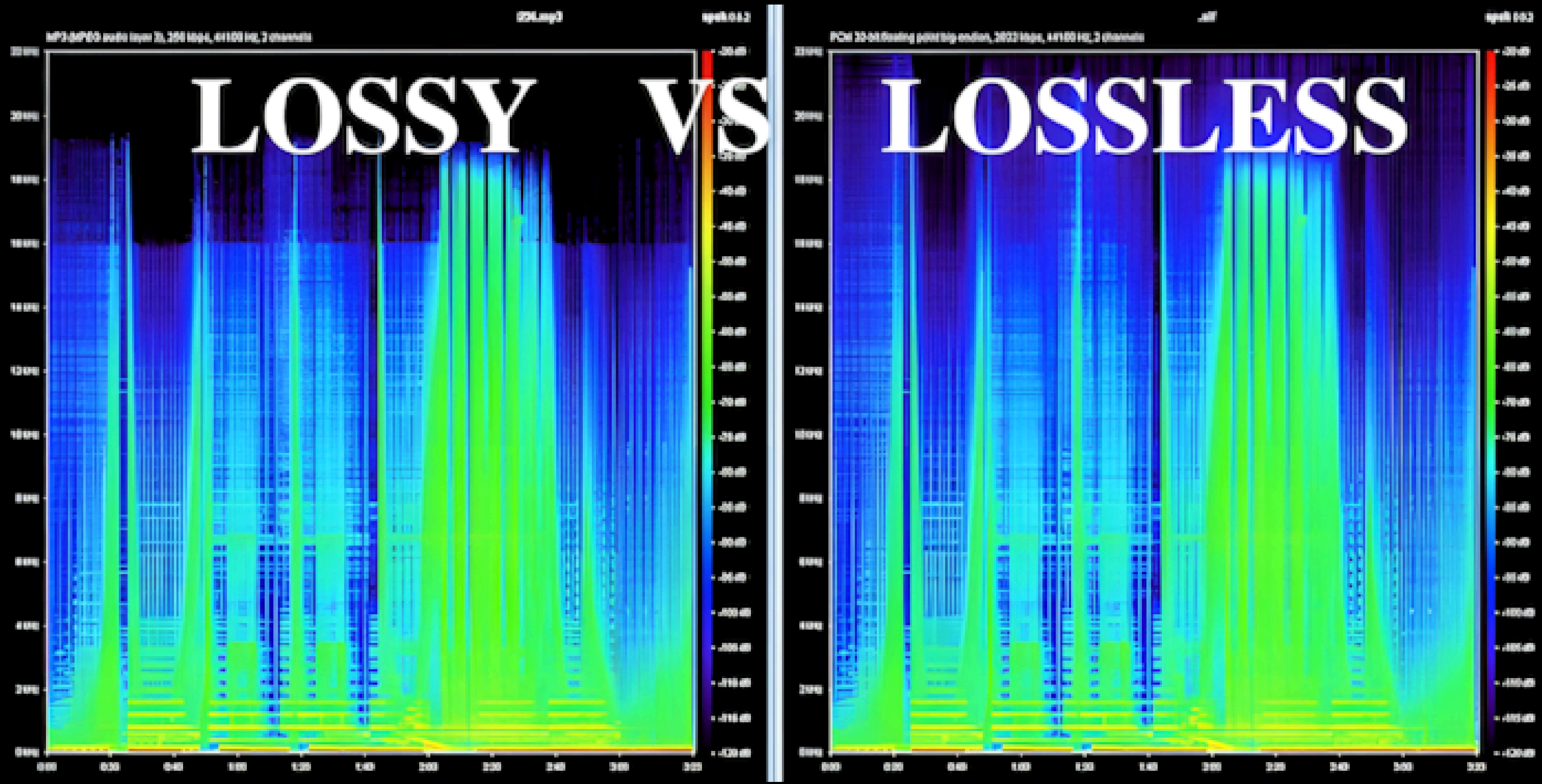
We naturally think of compression in terms of a transformation from a information that is less compressed to information that is more compressed.

But we also have to think about the inverse transformation: from information that is more compressed to less compressed (decompression).

When a method of compression produces a compressed output that can be decompressed deterministically into the original input, we have *lossless* compression.

Otherwise, the method of compression is *lossy*: some information from the original input cannot be recovered by decompression.


LOSSY VS LOSSLESS


LOSSY VS LOSSLESS

ANNALS OF ARTIFICIAL INTELLIGENCE

CHATGPT IS A BLURRY JPEG OF THE WEB

OpenAI's chatbot offers paraphrases, whereas Google offers quotes. Which do we prefer?

By Ted Chiang

February 9, 2023

COMPRESSION SCHEMES

There is an important principle concerning compression that is crucial to understand the function that it may play in cognition:

PIGEONHOLE PRINCIPLE if n items are put into m containers, with $n > m$, then at least one container must contain more than one item.

An important consequence: *no lossless compression scheme strictly compresses every input*. This holds even if we are dealing with 'optimal' compression schemes.

This gives us a sense of *incompressibility*. We usually deal with a laxer notion, which takes into consideration other tradeoffs, such as execution time and space. Another sense of incompressivility: the abscence of pattern. Compressing reduces patterns in representations; maximally compressed representations become incompressible.

The only way to (potentially) avoid this consequence is to adopt a lossy scheme or (inclusive) to adopt a split and conquer strategy: separate the input into bits each of which can be strictly compressed by potentially different compression schemes.

IDEALISATION AS A COMPRESSION STRATEGY

In either case, compression requires *specialization* (specialization requires understanding).

Idealisation is a compression *strategy* (or a family of such strategies): a process through which certain outcomes are achieved.

It is rationalized by the need to specialize representations in the context of ongoing tasks.

An idealisation is the result of using a compression scheme to encode our representation of a target.

IDEALISATION AS LOSSLESS COMPRESSION?

What kind of compression schemes are involved in idealisation?

Consider lossless compression: that would mean that it should be possible to turn idealised representations into de-idealised representation without loss of information.

This, of course, is not the case: from an idealised model, it is not in general possible to recover properties of the original model that have been idealised away.

The mapping from idealised models to external structure does not preserve uniqueness.

What we can do sometimes is to recover a minimal model from a family of models that converge in certain features (cf. Wimsatt 2007).

IDEALISATION AS LOSSY COMPRESSION

If idealisation involves compression, it must be *lossy*.

Take a system with a structure S. Produce a hypothesis H about S. Then, we can discard (part of) the difference between H and S. Whatever is discarded from S in the hypothesis generation and from the difference between S and H is idealised away. This results in compression with loss.

ABSTRACTION AND IDEALISATION

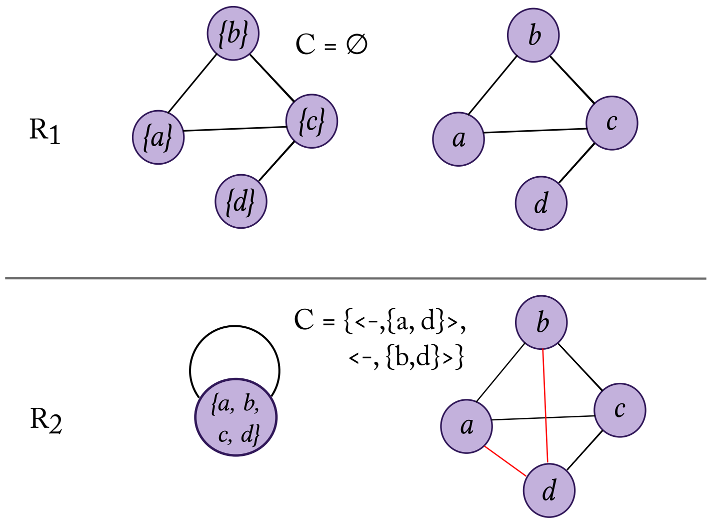
Frigg (2023) distinguishes between abstraction and idealisation:

- *abstraction* is ‘the wholesale omission of a property [...] the property is ommitted and [...] does not fall under a respect that is represented in the model’.
- *idealisation* is the distortion of a property that falls under a respect that is represented in the model.

Both are effects of compression: in abstraction, compression is achieved by discarding elements of what is represented, and in idealisation the scheme distorts in offering a more compressed representation of the target.

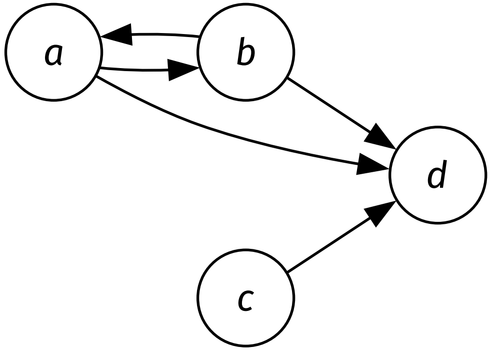
TRADEOFFS

Idealisation trades length of the hypothesis for length of the correction code.



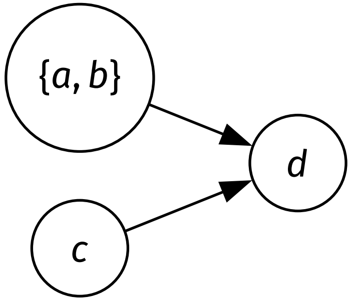
Interpret these graphs as concerning some relational structure. To say that parameters *a*...*d* are all connected is false, but idealises the model.

It suffices that the length of a *possible* correction is pessimised.



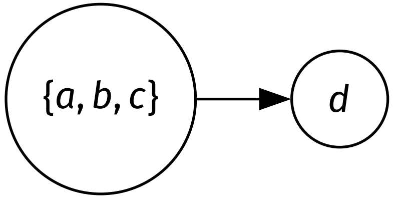
$C = \varnothing$

$$\left\langle \langle a, b, c, d \rangle, \varnothing, \begin{array}{|c|c|c|c|} \hline 0 & 1 & 0 & 1 \\ \hline 1 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 0 \\ \hline \end{array}, \varnothing \right\rangle$$



replace $\{a, b\}$ with its complete graph

$$\left\langle \langle a, b, c, d \rangle, \{\{a, b\}\}, \begin{array}{|c|c|c|} \hline 0 & 0 & 1 \\ \hline 0 & 0 & 1 \\ \hline 0 & 0 & 0 \\ \hline \end{array}, \varnothing \right\rangle$$



replace $\{a, b, c\}$ with its complete graph, disconnect c from a and b

$$\left\langle \langle a, b, c, d \rangle, \{\{a, b, c\}\}, \begin{array}{|c|c|} \hline 0 & 1 \\ \hline 0 & 0 \\ \hline \end{array}, \langle \varnothing, \{\{c, a\}, \{c, b\}\} \rangle \right\rangle$$

LIMIT IDEALISATIONS

Abstractive idealisations involve compression, but there are other kinds of idealisations.

Limit idealisations ‘push a certain property to the extreme’, describe what happens when a certain parameter is assigned or approaches a limit value.

In general, this involves compression only in a limited sense, by compression of the parameter.

An idea: maybe by pushing parameters to the limit, we obtain a compressed representation of the space of possible configurations (we do not tend to use limit idealisations on their own, but as special cases of a family of models).

MULTIPLE-MODELS IDEALISATION

In a multiple-model idealisation, multiple models are used to represent a target.

This is a reflection of the divide and conquer strategy I sketched above: if no scheme is available that is optimal for compressing a corpus, it is often viable to split the corpus and try different schemes for different parts.

Strictly speaking, this is more general than usual descriptions of multiple-models idealisation (cf. Weisberg 2013).

Compression might be a better justification for the use of model ensembles.

SUMMARY

Treating certain kinds of idealisation as kinds of compression strategies makes sense.

Idealisation helps with understanding insofar as compression helps with understanding.

Compression (as outcome) already requires understanding, in order to set up an appropriate compression scheme to use.

Different compression schemes yield different types of idealisation.

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Thanks!

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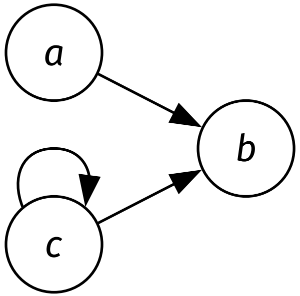
In advanced physics, you learn about perfectly spherical cows *under rotation*.

ADJACENCY MATRICES

A (di)graph G of n vertices can be represented by a $n \times n$ matrix m, where each cell $m_{i,j}$ represents whether the vertex i is connected to vertex j.



0	1	0
0	0	0
0	1	1

A graph is complete, if each vertex is connected to every other vertex.

The matrices of complete graphs alwaysys have adjacency matrices of the form

0	1	1	…
1	0	1	…
1	1	0	…
…	…	…	…

SUAREZ AND BOLINSKA ON MODELS AND INFORMATION

What I have presented here has some connections to work by Suárez & Bolinski (2021), who apply the tools of information theory to the case of abstraction and idealisation.

They consider models as a medium for communicating information about a target ("a tool to *compactly* convey codified information regarding the system of interest", p. 77, emphasis mine), and then they recognize that such mediums can be *noisy* and subject to *equivocation*.

They treat this as a helpful analogy, rather than a full description of idealisation and abstraction.

In the case of abstraction, S&B say that there is "equivocation", as some information available at the source qoes not get transmitted to the receiver (the user of the model).

In the case of idealisation, information is given that is not present in the source, and thus can be taken as 'noise'.

TED NELSON AGAINST COMPRESSION

> *It was an experience of water and interconnection [...] I was trailing my hand in the water and I thought about how the water was moving around my fingers, opening on one side and closing on the other. And that changing system of relationships where everything was kind of a similar and kind of the same, and yet different. That was so difficult to visualize and express, and just generalizing that to the entire universe, that the world is a system of ever-changing relationships and structures struck me as a vast truth. Which it is. [...] And writing is the process of reducing a tapestry of interconnections to a narrow sequence. And this is in a sense illicit. This is a wrongful compression of what should spread out.*
>
> — *Werner Herzog's* Lo and Behold